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#include<stdio.h>
#include<conio.h>
#include<string.h>
#include<stdlib.h>
#include<ctype.h>
char mnemonic[3][3][10]=
{
    {"1","START","AD"},
    {"2","EQU","AD"}
};
char symbol_table[10][2][10]={""};
int s1=0;
int main()
{
    int i=0,j=0;
    int loc=0;
    int start=0,equ=0,origin=0;
    char *field,record[200],const1[10];
    char symb_loc[25];
    int n;
    char op[20];
    FILE *fr;
    clrscr();

    fr=fopen("ass_2.txt","r");
    while(!feof(fr))
    {
        int fcnt=0;
        loc++;
        fgets(record,200,fr);
        field=strtok(record," ");

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while(field!=NULL)
{
    fcnt++;
    printf("%s \t",field);

    if(fcnt==1)
    {
        if(strcmp(field,"$")!=0)
        {
            strcpy(symbol_table[s1][0],field);
            strcpy(op,field);
            sprintf(symb_loc,"%d",loc);
            strcpy(symbol_table[s1][1],symb_loc);
            s1++;
        }
    }
    if(fcnt==2)
    {
        int found=0;
        int index;
        for(i=0;i<3;i++)
        {
            if(strcmp(mnemonic[i][1],field)==0)
            {
                found=1;
                index=i;
                break;
            }
        }
        if(found==1)

```

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{
    char class1[10]="";
    char mnemonic1[10]="";
    strcpy(class1,mnemonic[index][2]);
    strcpy(mnemonic1,mnemonic[index][1]);
    if(strcmp(class1,"AD")==0)
    {
        if(strcmp(mnemonic1,"START")==0)
        {
            start=1;
        }
        if(strcmp(mnemonic1,"EQU")==0)
        {
            equ=1;
            loc--;
        }
    }
}
}
if(fcmt==3)
{
    if(start==1)
    {
        strcpy(const1,field);
        loc=atoi(const1);
        loc=loc-1;
        start=0;
    }
    if(equ==1)
    {
        char index_of_symbol[20];

```

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int find_index=0;
for(i=0;i<s1;i++)
{
    if(strcmp(symbol_table[i][0],field)==0)
    {
        if(strcmp(symbol_table[i][1],"")!=0)
        {
            find_index=1;
            strcpy(index_of_symbol,symbol_table[i][1]);
            break;
        }
    }
}
if(find_index==1)
{
    for(i=0;i<s1;i++)
    {
        if(strcmp(symbol_table[i][0],op)==0)
        {
            strcpy(symbol_table[i][1],index_of_symbol);
            break;
        }
    }
    find_index=0;
}
equ=0;
}
}
field=strtok(NULL," ");
}
}

```

```
fclose(fr);

printf("\n \n \n symbol table\n");

for(i=0;i<s1;i++)
{
    printf("\n");
    for(j=0;j<2;j++)
    {
        printf("%s \t",symbol_table[i][j]);
    }
}
getch();
return 0;
}
```